

FEED GAS PARTICULATE FILTER

IFC	1		ISSUED FOR CONSTRUCTION	MTOK	MMAR	IC
IFA	0	24-Jun-22	ISSUED FOR APPROVAL	MTOK	MMAR	IC
Status	Rev. n.	Date	Description	Prepared	Checked	Approved
	Rev. Index					
Client  SOUND ENERGY Mahaj Riad Center, Batiment 6, 3eme etage, Avenue Al Arz, Hay Ryad, 10100 Rabat, Morocco				Project Title TENDRARA PROJECT		
Contractor  ITALFLUID GEOENERGY Via Antonio Vivaldi, 4 Montesilvano (PE) ITALY Tel:085-4224374 / fax: 085-4220742				Provision of a leased train for production of LNG for Sound Energy		
Engineering  ITALFLUID GEOENERGY Via Antonio Vivaldi, 4 Montesilvano (PE) ITALY Tel:085-4224374 / fax: 085-4220742						
Technology Vendor   TECHNICAL AMERICA A Technical Group Company				TECHNICAL AMERICA INC 301 N Smith Avenue, 92880 Corona (CA)-USA Ph. +1 951 272 9540		
Facility Name TENDRARA			Location MOROCCO	Scale A4	Sheet of Sheets 1 of 4	
Document Title FEED GAS PARTICULATE FILTER				Doc. Code PRO11-ING-FR02		
				Vendor Doc. code 21044-PR-DSH-0F201		
Software: Microsoft Excel				File name: PRO11-ING-FR02		

Technology vendor		DOC TITLE		FEED GAS PARTICULATE FILTER		REV	
		TG DOC NUMBER		21044-PR-DSH-0F201		1	
Customer/ Engineering		ITF DOC NUMBER		PRO11-ING-FR02		PAGE	
		PROJECT		TENDRARA MICRO LNG PLANT		2	

1	Plant/Process Unit Name	LNG Tendrara plant / UNIT 200		Location	Tendrara		
2	Equipment Service	Feed gas particulate filter		Quantity of Units	2		
3	Equipment TAG	F-201A/B		P&ID	PRO13-ING-PR01		
4	Applicable to	☐ Proposal ☒ Construction		As Built	☐ As Built		
5	Model/Size and Type	Feed gas particulate filter		Purchaser/Agent	TG		
6				Manufacturer/Supplier	CALORIFER		
7							

GENERAL							
9	Size	Vertical		Volume:	0,27	m3	
10	Orientation	18" DIA mm		Empty Weight:	TBD	kg	
11	Dimension	1625 sm to sm mm		Hyd. Test Weight:	TBD	kg	
12	Thickness	12,7 mm (SCH80S)		Operating Weight:	TBD	kg	
13							
14							

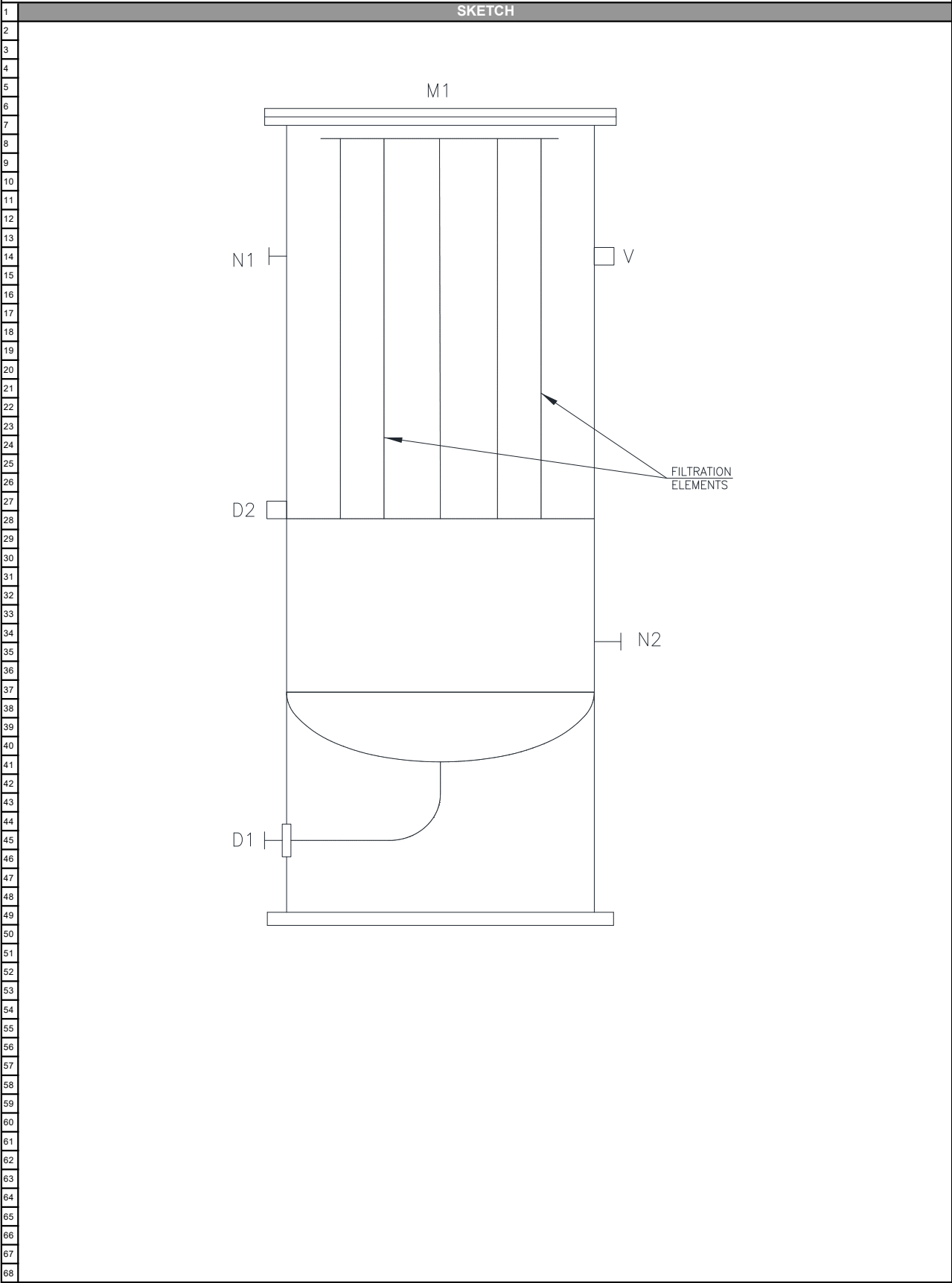
OPERATING CONDITIONS							
16	Operating Pressure:	40,0	barg	Liquid Filtered:	Fluid	Hydrocarbon/Water	
17	Operating Temperature:	9,7	°C		Flow Rate	477,89	kg/h
18	Gas	Fluid	Natural gas		Molecular Weight	121,00	
19		S. Vol. Flow Rate	509.420,1		Density (mix)	527,4	kg/m3
20		Molecular Weight	19,8		Viscosity (max)	1,11	cP
21		Density:	39,8				
22		Viscosity	0,0123				
23							

DESIGN CONDITIONS							
25	Design Pressure:	50,0	barg @	100	°C	Radiography required:	
26	MDMT	-28,0	°C @	50,0	barg	☐ Spot ☒ Legs / Lifting Lugs	
27	Corrosion Allowance:	0,0	mm			Accesories: YES Internals: YES	
28	Design Code:	ASME Sec VIII, Div. 1				Type: SS304 particulate catridge	
29	Code Stamp Required:	NO				(5 micron rating)	
30	Painting:	Yes - Note 3				Qty & Cartidge P/N: xx	
31	Insulation:	NO				PWHT: No	
32						Test Pressure (Hydro): 65 barg	
33							

CONSTRUCTION MATERIALS							
34	Shell	SA312 TP304				Bolting: SA 193 - B7 /SA 194 -2H FRP coated	
35	Nozzles	SA312 TP304				Gaskets: 304 inner/outer ring Graphite/304 SPWD	
36	Flanges	SA-182 - F304/304L					
37	Pipes	SA-312 TP304					
38	Legs	SA 240 304 / S235 JR					
39	Lifting lug	SA 240 304					
40							

NOZZLES							
42	Tag	Qty	Día (in)	Type	Rating	Sch.	Service
43	N1	1	6	WNRF	#600	40S	Inlet
44	N2	1	6	WNRF	#600	40S	Gas out
45	V	1	1/2	NPT	#6000	-	Vent
46	D1	1	1	WNRF	#600	40S	Drain
47	D2	1	1/2	NPT	#6000	-	Drain
48	M1	1	18	WNRF	#600	80S	Manway
49							
50							
51							
52							
53							
54							
55							
56							
57							
58							
59							
60							
61							
62							
63							
64							
65							
66							
67							
68							

Technology vendor 	DOC TITLE	FEED GAS PARTICULATE FILTER	REV
	TG DOC NUMBER	21044-PR-DSH-0F201	1
	ITF DOC NUMBER	PRO11-ING-FR02	PAGE
Customer/ Engineering 	PROJECT	TENDRARA MICRO LNG PLANT	3



Technology vendor  	DOC TITLE	FEED GAS PARTICULATE FILTER	REV
	TG DOC NUMBER	21044-PR-DSH-0F201	1
Customer/ Engineering 	ITF DOC NUMBER	PRO11-ING-FR02	PAGE
	PROJECT	TENDRARA MICRO LNG PLANT	4

1	NOTES
2	
3	1 - Test pressure shall be witnessed by authorized Moroccan inspector to allow importation in Morocco.
4	2 - Deleted
5	3 - Refer to painting procedure MEC07-ING-FR03 (21044-QC-PAI-00001).
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	
16	
17	
18	
19	
20	
21	
22	
23	
24	
25	
26	
27	
28	
29	
30	
31	
32	
33	
34	
35	
36	
37	
38	
39	
40	
41	
42	
43	
44	
45	
46	
47	
48	
49	
50	
51	
52	
53	
54	
55	
56	
57	
58	
59	
60	
61	
62	
63	
64	
65	
66	
67	
68	